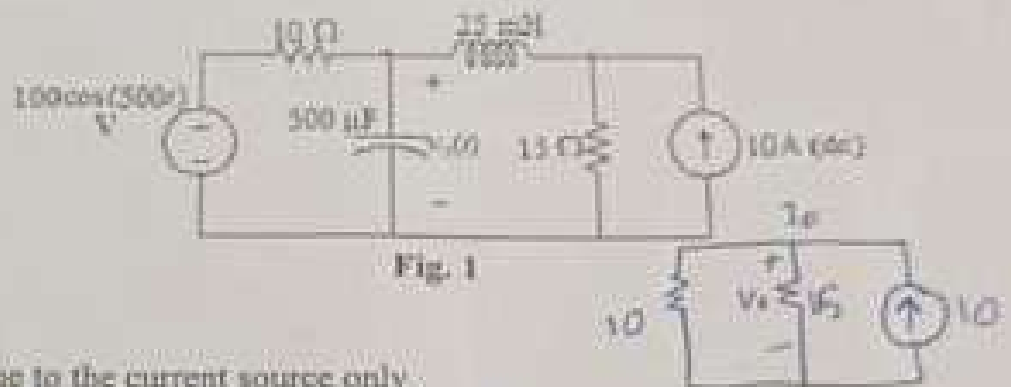


With all the required assumptions (without simplification) to solve the following problems, find and find the final answers.

Prob.1: Consider a source where a voltage source $v(t) = 100\cos(500t + 10^\circ)$ supplies a by a current $i(t) = 1\sin(500t + 80^\circ)$.

1. The equivalent impedance at the terminals of the voltage source is $Z = 50 \angle 50^\circ$
2. The current leads the voltage by 50°
3. The voltage leads the current by 50°
4. The average value P_{avg} at $t = 100\mu s$ is 7.234
5. The phasor domain representation of the voltage is $50 \angle 130^\circ$
6. The frequency of the current is 157.08
7. Draw the phasor diagram for both $v(t)$ and $i(t)$

Prob.2: Consider the circuit shown in Figure 1 use superposition to find $v(t)$. Note that the current source is dc.



a. The voltage V_0 due to the current source only

$$I_s = \frac{10 \times 10}{10 \times 15} = 0.666 A \quad V_0 = I_s \times R = 0.66 \times 15 = 10$$

$$V_0 = 10 V$$

b. The voltage V_0 due to the voltage source only

$$V_0 = \frac{100 + -4j}{10 - 4j} = 37.139 \angle -68.19^\circ = 37.139 - 34.48j$$



c. The value of V_0 due to both sources

$$V_0 = V_0' + V_0'' = 10 + 37.139 - 34j$$

$$V_0 = 41.82$$

$$23.79 - 34j$$

Prob.3: Consider the circuit shown in Figure 2.

1. The Thevenin voltage (V_{th}) with respect to terminals a-b of the circuit.

$$100 - j5 = 2 - j4$$

$$V = I Z = 89.37 \angle -33.4^\circ$$

$$I = \frac{V}{Z} = \frac{89.37 \angle -33.4^\circ}{2 + j2} = 31.6 \angle -73.4^\circ$$

$$I_n = 31.6 \angle -73.4^\circ$$

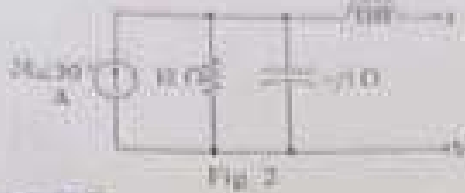


Fig 2



$$V_o = V_{th} = I Z = (31.6 \angle -73.4^\circ)(2 + j2) = 89.37 \angle -33.4^\circ$$

2. The Norton current (I_n) with respect to terminals a-b of the circuit.

$$I_n = 31.6 \angle -73.4^\circ$$

3. The Thevenin equivalent impedance between the a-b terminals

$$Z_{th} = \frac{V_{th}}{I_n} = \frac{89.37 \angle -33.4^\circ}{31.6 \angle -73.4^\circ} = 2 + j2 \Omega$$

4. Draw the Thevenin equivalent circuit, then find the load current when $Z_L = 3 - j4 \Omega$ is connected between the a-b terminals

$$V = I Z$$

$$I = 17.87 \angle 49.73^\circ$$

$$89.37/33 = I(3 - j4)$$

Prob. 4: Consider the circuit shown in Figure 3.

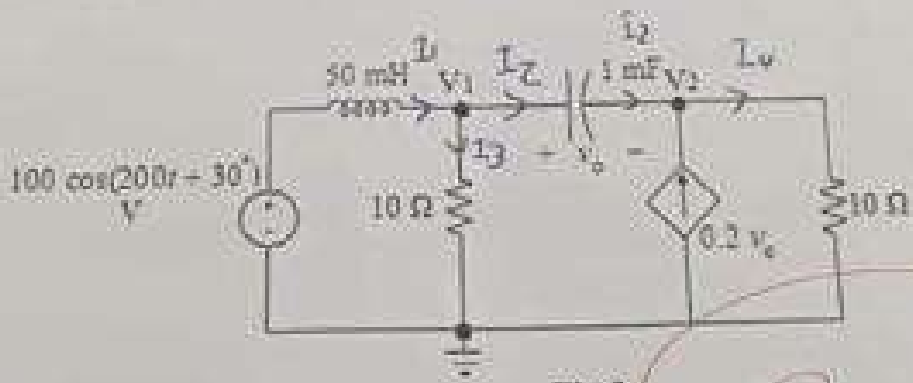


Fig 3

Using the nodal method, write the required equations that you are needed to find V_1 , V_2 , and V_0

$$\frac{100 \angle 30^\circ}{-j5} = \frac{V_1 - V_2}{-5j} + \frac{V_1}{10} \Rightarrow 10 \angle 30^\circ = (1 + 2j)V_1 - 2jV_2$$

$$12V_1 - 17V_2 = 0$$

Good luck